

https://www.youtube.com/watch?v=Sb1JIDUw3kM
Scope

A) Motor controller for at least one wheel, goal is just motor spinning open loop
B) If we're goated: bms, battery, escs, controller
C) Kicad, reflow
D) ~4 hrs each a week for 4 weeks, same pace as labs
E) Have cells, need nickel strips + spot welder (ask GR but prob need to buy our own if make battery)
F) Ask ieee for $
G) Proposal due end of week

Motor Controller
Week 1
Spec Components & Research

- **Microcontroller (MCU):** the chip, crystal oscillator, reset circuit, and local decoupling capacitors). **STM32H743 seems like a pretty good chip/popular/well-suported and good longterm for the board**
    - Using Silicon Oscillator so need to run in **HSE Bypass mode**
    - Make sure traces running PWM signal from STM32 to 3 phase gate driver are as short as possible
    - PA0 will be emergency shutoff for the gate drivers, making it low will cause shutoff, should be normally high
- **Power Supply (Buck/LDOs):** switching regulator IC, inductor, diodes, and filter caps to drop battery voltage down to 12V and 3.3V
    - MP9486 to get 12V rail.
- **Gate Driver:** gate driver IC, bootstrap diodes/capacitors, and gate resistors).
    - 3x IRS2001 from digikey
    - Must **configure dead time** so it doesn't explode (Some do it in hardware but this one doesnt)
- **Power Stage (MOSFETs):** 6 high-power N-channel MOSFETs, gate pull-down resistors, and snubber components).
    - Want very low R_{DS} for power mosfets to lower heat generation
- **Bulk Capacitors:** low-ESR electrolytic capacitors to absorb voltage spikes).
- **Connectors & Interfaces:** battery connector, 3 motor phase terminals, USB programming port, and receiver/throttle header).
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STM32

Week 2
Schematics
PCB Design

Week 3
PCB Design
LT Spice
Order Components 2 days

Board 3-5 bis days

Week 4
Put the shit in the bag